The scaling of forced collisionless reconnection

Brian P. Sullivan[a] and Barrett N. Rogers
Department of Physics and Astronomy, Dartmouth College, Hanover, New Hampshire 03755

M. A. Shay
Institute for Research in Electronics and Applied Physics, University of Maryland, College Park, Maryland 20742

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We present two-fluid simulations of forced magnetic reconnection with finite electron inertia in a collisionless two-dimensional slab geometry. Reconnection in this system is driven by a spatially localized forcing function that is added to the ion momentum equation inside the computational domain. The resulting forced reconnection process is studied as a function of the temporal and spatial structure of the forcing function, the plasma β, and strength of the out-of-plane guide magnetic field component, and the electron to ion mass ratio. Consistent with previous results found in unforced, large Δ' systems, for sufficiently strong forcing the reconnection process is found to become Alfvénic, i.e., the inflow velocity scales roughly like some small fraction of the Alfvén speed based on the reconnecting component of the magnetic field just upstream of the dissipation region. The magnitude of this field and thus the rate of reconnection is controlled by the behavior of the forcing function. When the forcing strength is below a certain level, fast reconnection is not observed. © *2005 American Institute of Physics*. [DOI: 10.1063/1.2146910]

I. INTRODUCTION

Magnetic reconnection is a ubiquitous process in which magnetic field lines embedded in a plasma break and reform, releasing large amounts of energy in the form of plasma flows and particle heating. The rate of reconnection is of key importance in determining if a particular model of reconnection is consistent with physical systems. Simulations based on traditional resistive magnetohydrodynamic (MHD) models, for example, predict relatively slow rates of reconnection due to the formation of long, narrow current layers[1,2] of the Sweet-Parker type.[3,4] Physics models that go beyond resistive MHD and include the Hall term in the generalized Ohm's law, on the other hand, typically predict the formation of open, Petschek-type[5] magnetic configurations and substantially faster rates of reconnection.[6–14] In the case of systems with narrow current sheets and large positive Δ', for example, several studies have found that the reconnection process in the presence of the Hall term becomes Alfvénic—that is, the inflow velocity scales roughly like some small fraction ($\sim 1/10$) of the Alfvén speed based on the reconnecting component of the magnetic field just upstream of the dissipation region, regardless of the system size[12–14] or the dissipation mechanism.[6,15–17] Various other scalings of the reconnection rate have also been reported in studies based on somewhat different physics models, system configurations, or parameter regimes.[18–22]

Magnetic reconnection in a given system is traditionally categorized as either spontaneous or forced. In the spontaneous case, reconnection arises from a linear instability in the system, while in the forced case, reconnection is driven predominantly by some externally applied force. Here, we explore the scaling of magnetic reconnection in a simple system of the forced type using two-dimensional, two-fluid numerical simulations with finite electron inertia. We consider a slab plasma geometry with a sinusoidally varying magnetic field component, an optional guide magnetic field component, and periodic boundary conditions. The configuration is designed to satisfy the tearing mode stability condition ($\Delta' \leqslant 0$) for all wavelengths allowed in the simulation and therefore, in the absence of forcing, no reconnection is observed to occur. Rather, reconnection in our system is driven entirely by the external force. Such a force is often imposed in the form of a finite amplitude perturbation on the walls of the simulation box.[23–27] In this paper, following a different approach, reconnection is driven by a spatially localized forcing function that is added to the ion momentum equation in the interior of the simulation domain. This function represents a generic external forcing agent that drives plasma and magnetic flux toward a predetermined reconnection region at a controllable rate and in a spatially controllable way. We investigate the behavior of the reconnection as a function of various free parameters in the system, including the temporal and spatial structure of the forcing function, the plasma β, and the presence of an out-of-plane guide magnetic field component.

Consistent with previous studies of spontaneous reconnection in systems with narrow current sheets (e.g., Ref. 6), we find that for sufficiently strong forcing the reconnection process becomes Alfvénic. In other words, as noted above, the inflow velocity is proportional to the Alfvén speed based on the reconnecting magnetic field just upstream of the dissipation region. As demonstrated in previous studies (e.g., Ref. 28) and as discussed later here, this scaling results in a reconnection rate that increases in proportion to the *square* of the reconnecting magnetic field strength. In contrast to spontaneously reconnecting systems, the magnitude of this up-

[a]Electronic mail: brian.p.sullivan@dartmouth.edu

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stream field in our system and thus the rate of reconnection is ultimately controlled by the strength and geometry of the forcing function. This function produces a pileup of magnetic flux in the upstream region and hence an increase in the upstream magnetic field, until the reconnection rate becomes sufficient to prevent further pileup of magnetic flux. In our simulations, this pileup typically halts when a relatively thin current sheet is formed—about half an ion skin depth in width—at which point the magnetic separatrix is observed to open up and fast (Alfvénic) reconnection onsets. Conversely, if the forcing strength is below a certain level, we find that the system settles into a new quasiequilibrium before a narrow current sheet is formed, and the resulting reconnection rate is much slower.

The organization of this paper is as follows. In Sec. II we describe the simulation model, equilibrium profiles, and forcing function. In Secs. III and IV we discuss the results of the nine simulations included in this study. The main conclusions are summarized in Sec. V.

II. SIMULATION MODEL

Our simulations are based on the two-fluid equations with the addition of a vertical forcing term in the ion momentum equation. These equations in normalized form are[28]

$$n(\partial_t \mathbf{V}_i + \mathbf{V}_i \cdot \nabla \mathbf{V}_i) = \mathbf{J} \times \mathbf{B} - \nabla p + F_y \hat{\mathbf{y}}, \tag{1}$$

$$\partial_t \mathbf{B}' = -\nabla \times \mathbf{E}', \tag{2}$$

$$\mathbf{E}' + \mathbf{V}_i \times \mathbf{B} = \frac{1}{n}(\mathbf{J} \times \mathbf{B}' - \nabla p_e), \tag{3}$$

$$\partial_t n + \mathbf{V}_i \cdot \nabla n = -n \nabla \cdot \mathbf{V}_i, \tag{4}$$

$$p_e = nT_e, \quad p_i = nT_i, \quad p = p_i + p_e, \tag{5}$$

$$\mathbf{B}' = \left(1 - \frac{m_e}{m_i}\nabla^2\right)\mathbf{B}, \quad \mathbf{V}_e = \mathbf{V}_i - \mathbf{J}/n, \quad \mathbf{J} = \nabla \times \mathbf{B}. \tag{6}$$

Here, $F_y = F_y(x,y,t)$ is the forcing function, the form of which is discussed in detail below. For simplicity we assume an isothermal equation of state for both electrons and ions (qualitatively the same as the $\gamma = 5/3$ case), and thus take T_e and T_i to be constant. The normalizations of Eqs. (1)–(6) are based on constant reference values of the density n_0 and the reconnecting component of the magnetic field B_{x0}, and are given by (normalized $\rightarrow$ physical units): $t \rightarrow \omega_{ci} t$, $\omega_{ci} = eB_{x0}/(m_i c)$, $x \rightarrow x/d_i$, $d_{i,e} = c/\omega_{pi,e}$, $\omega_{pi,e}^2 = 4\pi n_0 e^2/m_{i,e}$, $n \rightarrow n/n_0$, $\mathbf{B} \rightarrow \mathbf{B}/B_{x0}$, $\mathbf{V}_{i,e} \rightarrow \mathbf{V}_{i,e}/V_{Ax}$, $V_{Ax} = \omega_{ci} d_i = B_{x0}/(4\pi n_0 m_i)^{1/2}$, $T_{i,e} \rightarrow T_{i,e} 4\pi n_0/B_{x0}^2$, $p_{i,e} \rightarrow p_{i,e} 4\pi/B_{x0}^2$, $\mathbf{J} \rightarrow \mathbf{J}/(n_0 e V_{Ax})$, and $F_y \rightarrow F_y d_i/(m_i n_0 V_{Ax}^2)$. Our algorithm employs fourth-order spatial finite differencing and the time-stepping scheme is a second-order accurate trapezoidal leapfrog.[29,30] We consider a square 2D simulation box with physical dimensions $L \times L = 102.4 d_i \times 102.4 d_i$ (so that $L = 102.4$ in normalized units) and periodic boundary conditions imposed at $x = \pm L/2$ and $y = \pm L/2$. The simulation grid is $n_x \times n_y = 512 \times 1024$, yielding grid scales of $\Delta_x = 0.2$ and $\Delta_y = 0.1$. The electron to ion mass ratio is typically $m_e/m_i = 1/25$ so that the normalized electron skin depth is $d_e = \sqrt{m_e/m_i} = 0.2$. Except in the immediate vicinity of X points, the frozen-in law for electrons in this model is broken by the presence of finite electron inertia, which is manifested by the m_e terms in the definition of B' in Eq. (6). Very near the X points, however, the electron inertia terms in the simple model considered here become weak in quasisteady conditions, since both ∂_t and $\mathbf{v} \cdot \nabla$ become small (see, e.g., Ref. 31 for further discussion). The frozen-in law in these small regions is mainly broken in the simulations by the presence of numerical diffusion. Past studies[31] have shown that the resulting rates of reconnection are not sensitive to this level of diffision, and are in reasonable agreement with the rates obtained from particle simulations.[6]

A. Initial equilibrium

The normalized magnetic field and density profiles in our initial equilibrium are given by

$$\mathbf{B} = \sin\left[\frac{2\pi}{L}\left(y + \frac{L}{4}\right)\right]\hat{\mathbf{x}} + \frac{B_{z0}}{B_{x0}}\hat{\mathbf{z}}, \tag{7}$$

$$n = 1 + \frac{1 - B_x^2}{2(T_i + T_e)}. \tag{8}$$

As required by the boundary conditions, B is periodic under $y \rightarrow y + L$. Reconnection can occur along the lines $y = \pm L/4$, where the x component of the magnetic field vanishes. Note that the normalization parameter B_{x0} has been chosen so that the peak value of this initial field is unity. From the given form of n, it is apparent that $n = 1$ at this location, or in physical units, $n = n_0$. The density profile is chosen to satisfy the total pressure balance condition, which in normalized form is given by

$$n(T_i + T_e) + \frac{1}{2}(B_x^2 + B_z^2) = \text{const.} \tag{9}$$

Unless otherwise stated, we take the (constant) total temperature to be $T_{\text{tot}} = T_i + T_e = 1.0$, or in physical units $4\pi n_0 T_{\text{tot}}/B_{x0}^2 = 1$, so that the plasma β based on the reconnecting field has a (minimum) value of 2 where $B_x = 1$ and $n = 1$. The density reaches a maximum value of $n = 1.5$ in the center of the out-of-plane current sheets ($y = \pm L/4$) where $B_x = 0$. The initial current is carried by the electrons, and the ions are initially at rest. To prevent energy buildup at the grid scale, the simulations include fourth-order dissipation in the density and momentum equations of the form $\mu_4 \nabla^4$, where $\mu_4 = 5.1 \times 10^{-5}$.

We seed the system with a small initial magnetic field perturbation given by $\tilde{\mathbf{B}} = \hat{z} \times \nabla \tilde{\psi}$, where

$$\tilde{\psi} = b_0 \frac{L}{4\pi}\left\{1 + \cos\left[\left(\frac{4\pi y}{L} + \pi\right)\right]\right\}\sin\left(\frac{2\pi x}{L}\right), \tag{10}$$

and $b_0 = 0.002$ is a constant parameter. The sign of $\tilde{\psi}$ is chosen to produce x lines at $(L/4, -L/4)$, $(-L/4, L/4)$ and o lines at $(L/4, L/4)$, $(-L/4, -L/4)$. From this perturbation, an approximate initial value for the island width is $w = (2/\pi)\sqrt{L^2 b_0} \sim 2.91$.[28] To avoid physically artificial effects

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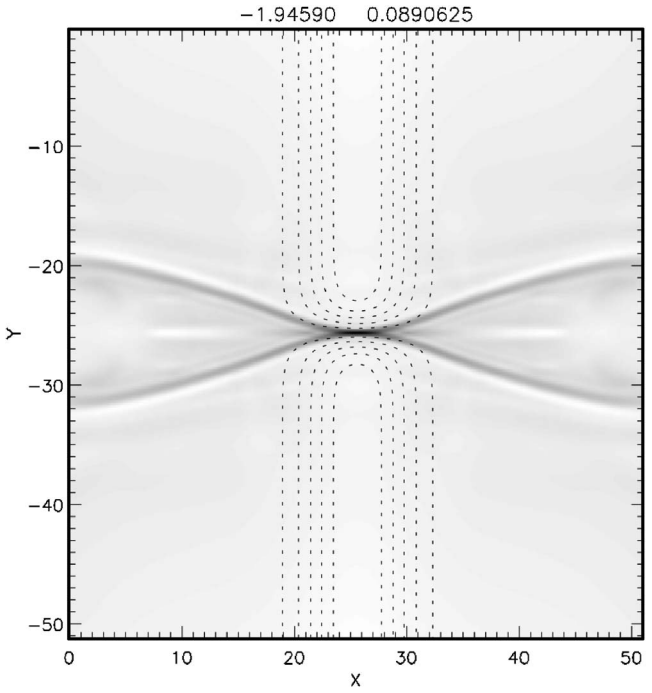


FIG. 1. $F_y(x,y)$ (dashed contours) and J_z (gray scale). The most negative current is shown in black. Maximum and minimum values of J_z are displayed above the figure. (Note: this figure depicts only one quarter of the simulation domain.)

that can arise from exact reflection symmetries of the initial condition (e.g., the trapping of a secondary magnetic island in the center of the current layer), a small amount of random noise is added to the magnetic field and ion current at the levels $|\tilde{B}_{\max}| \approx 10^{-4}$, $|\tilde{J}_{i\,\max}| \approx 10^{-4}$.

The linear tearing mode stability parameter is defined as $\Delta' = [\partial_y \tilde{\psi}(y \to 0+) - \partial_y \tilde{\psi}(y \to 0-)]/\tilde{\psi}(y=0)$, where $\tilde{\psi}$ is the perturbation in the flux function.[32] In our system it is given by[33]

$$\Delta'(k_x) = -\left(2k_0\sqrt{\frac{k_x^2}{k_0^2}-1}\right)\tanh\left(\sqrt{\frac{k_x^2}{k_0^2}-1}\right), \tag{11}$$

where $k_0 = 2\pi/L$. Since periodicity along the x direction requires $k_x \geqslant 2\pi/L$ and thus $k_x^2/k_0^2 \geqslant 1$, one sees that $\Delta' \leqslant 0$. Therefore, as noted in the Introduction, the system we consider is stable to tearing modes in the absence of forcing and exhibits no spontaneous reconnection.

B. Forcing function

The forcing function is designed to drive plasma and magnetic flux into the x line at $(x,y) = (L/4, -L/4)$ in the lower right-hand quadrant of the simulation. The shape of the function in a typical nonlinear simulation is shown in Fig. 1. This figure depicts only one quarter of the total simulation domain. This function has the general form $F_y(x,y,t) = X(x)Y(y)\Theta(t)$. Along x, the forcing function is a Gaussian of width w_x centered at $x = +L/4$

TABLE I. Simulation parameters.

Fig. no.	F_∞	w_x	$T_{\rm tot}$	B_{z0}
3	0.11	10	1.0	0.0
4	0.05	10	1.0	0.0
5	0.03	10	1.0	0.0
6	0.005	10	1.0	0.0
7	⋯	10	1.0	0.0
8	0.05	10	1.0	$5.0B_{z0}$
9	0.05	$10\sqrt{2}$	1.0	0.0
10	0.05	$10/\sqrt{2}$	1.0	0.0
12	0.05	10	0.5	0.0

$$X(x) = \exp\left[-\left(\frac{(x-L/4)}{w_x/2}\right)^2\right]. \tag{12}$$

Along the inflow (y) direction, the forcing function is antisymmetric in y about $y=-L/4$. It varies linearly with y close to the reconnection point, and then levels off to a constant value ($Y \to \pm 1$) over a distance of $\sim 2w_y$ upstream of the x point

$$Y(y) = \tanh\left(\frac{y-L/4}{w_y}\right) - \tanh\left(\frac{y+L/4}{w_y}\right) + 1. \tag{13}$$

In physical units $w_y = 2.22d_i$ in all the simulations included here. The impact of varying w_y in the simulations is discussed briefly in a later section.

The time behavior of the forcing is controlled by the function $\Theta(t)$. This function starts at zero and increases monotonically with time at the rate $d\Theta/dt \sim 1/\tau_f = 0.1$ until it plateaus at the value F_∞:

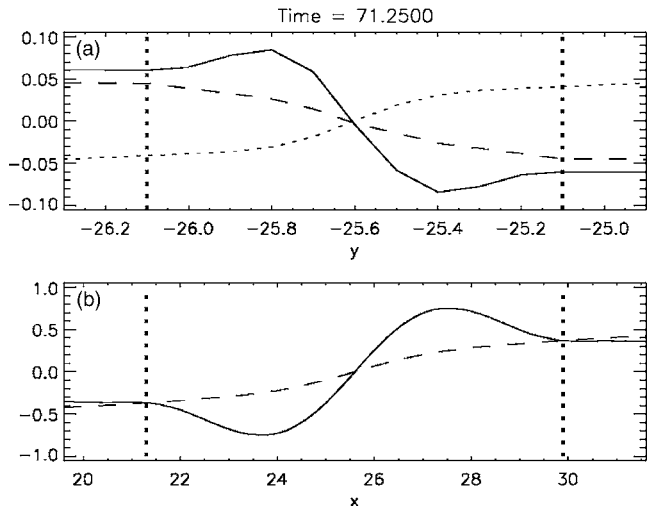


FIG. 2. Dissipation region parameters. (a) Inflow quantities: V_{ey} (solid), V_{iy} (dashed), $B_x(y)/10$ (dotted). (b) Outflow velocities: V_{ex} (solid), V_{ix} (dashed). The vertical dotted lines indicate the location at which inflow and outflow quantities are measured as described in the text. (These data are from the run shown in Fig. 4.)

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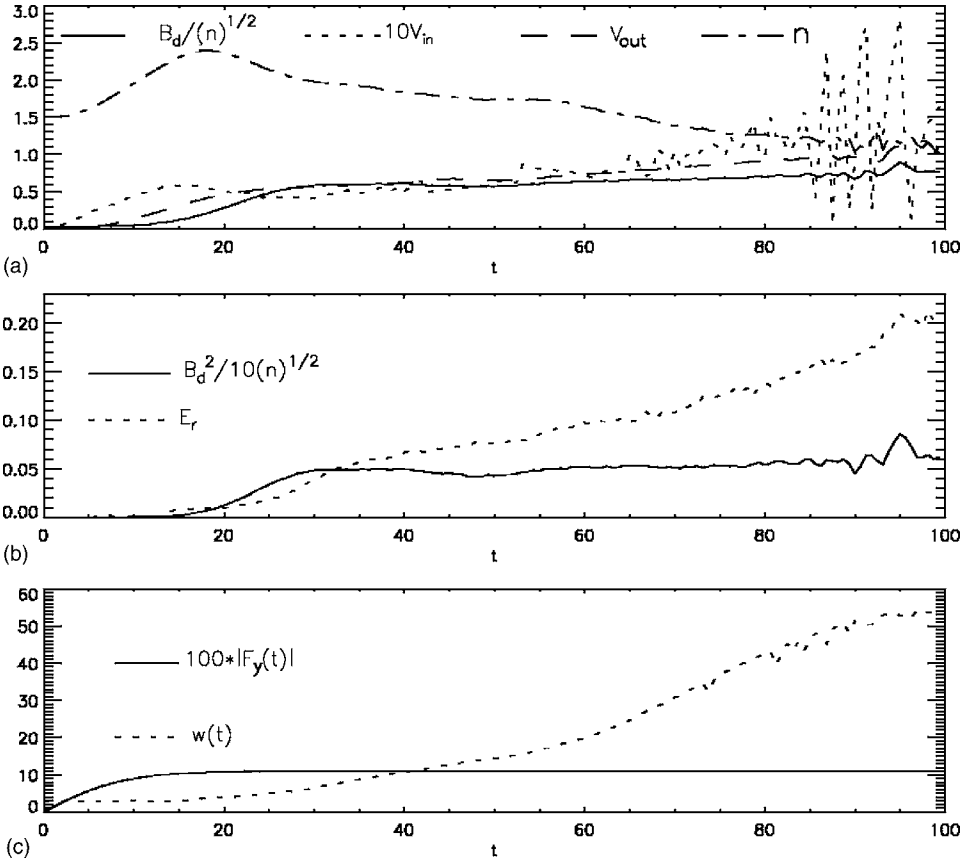


FIG. 3. Plateaued forcing with $F_\infty = 0.11$.

$$\Theta(t) = F_\infty \tanh\left(\frac{t}{|F_\infty|\tau_f}\right). \tag{14}$$

Since the initial slope of $\Theta(t)$ and hence the initial rate of ramping is held constant at $1/\tau_f = 0.1$ from one simulation to the next, it takes more time to ramp up to the stronger levels of forcing (roughly $\tau \sim |F_\infty|\tau_f$). In general, however, the ramping phase is short compared to the duration of the run and is completed before significant island growth begins.

III. RESULTS

Table I summarizes the key parameters of the nine simulations included in this study. The figure number for each simulation is shown in the left-hand column, followed by the asymptotic level of forcing F_∞, the horizontal width of the forced region w_x, the total temperature $T_{\text{tot}} = T_i + T_e$, and the equilibrium out-of-plane field B_{z0}. The first four simulations differ only in the asymptotic level of forcing F_∞. The final five simulations include one in which the magnitude of the forcing function is linearly ramped continuously for the entire simulation (Fig. 7), one simulation with a guide magnetic field of $B_{z0} = 5.0 B_{x0}$ (Fig. 8), two in which the width w_x of the forcing function has been widened and narrowed (Figs. 9 and 10), and one with a lower plasma-β than the others (Fig. 12).

Following Ref. 28, we focus here on the magnitudes of the ion inflow $V_{\text{in}}(=V_y)$, outflow $V_{\text{out}}(=V_x)$, and reconnecting magnetic field $B_d(=B_x)$ near the boundaries of the so-called ion dissipation region where the ions decouple from both the electrons and the magnetic field. In our zero-guide field simulations the ion and electron velocities tend to diverge approximately $0.5c/\omega_{pi}$ upstream of the x point. This location is not only the same from simulation to simulation but also is very stably fixed for the duration of a given simulation. We therefore measure the upstream quantities, V_{in} and B_d, at a point 0.5 ion skin depth directly upstream of the x point. However, varying this location between 0.5 and 1.0 produces relatively small quantitative changes in the figures shown in this paper, as these upstream quantities vary only weakly along y in this range as seen in Fig. 2(a). The position of the downstream edge of the dissipation region is more variable. As noted in previous studies, the electron outflow (in the x direction) very close to the x point typically becomes quite large and can greatly exceed the ion Alfvén speed. As the electrons approach the downstream edge of the dissipation region (typically about $\Delta x = D = 5$ downstream from the x point), however, they slow down to flow roughly with the ions [see Fig. 2(b)]. Following Shay *et al.*, the outflow V_{out} is therefore evaluated at the point where the electron and ion velocities first become equal downstream of the x point. Note that a typical value for the aspect ratio of the ion dissipation region is $\delta/D \sim 0.5/5 \sim 0.1$, in agreement with previous studies (e.g., Ref. 28).

The reconnection rate is defined as the time rate of change of flux through the dissipation region, and is measured by taking the time derivative of the difference in flux between the x point and the o point in the reconnecting current sheet, $E_r \equiv d\psi/dt$.

A. Plateaued forcing

Data from the first simulation are shown in Fig. 3. Time

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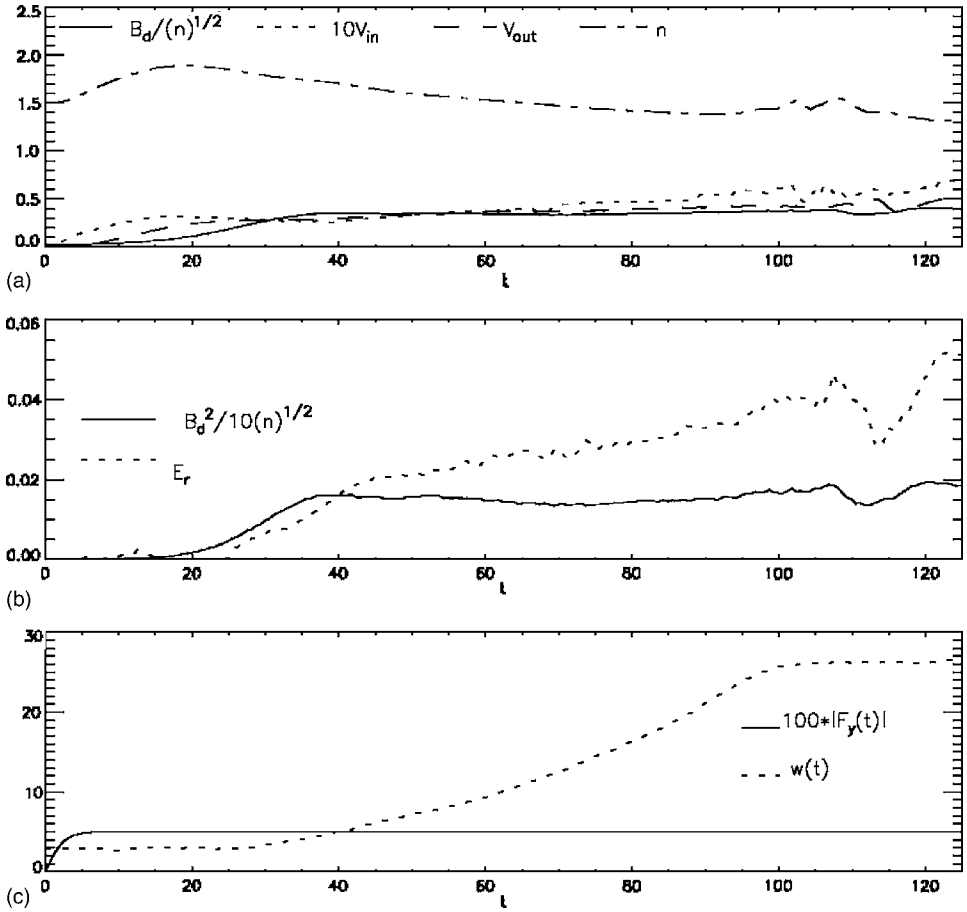


FIG. 4. Plateaued forcing with $F_\infty = 0.05$.

series of the magnitude of the forcing function and the width of the magnetic island are plotted in Fig. 3(c). Note that the level of forcing becomes constant by about $t=15$ before appreciable island growth has occurred, and that subsequently the island width grows for much of the simulation before finally leveling off at about half the system size. Also note that the reconnection rate plotted in Fig. 3(b) remains finite even after the island width has ceased to grow, indicating that the magnetic field inside the island is becoming stronger with time.

With one exception (the case of weakest forcing, Fig. 6) several features of this run are generic to all of the simulations presented here. Notice, for example, the hump in the inflow velocity (dotted line) during the time $t<30$ in Fig. 4(a). This initial increase in V_{in} is due to the ramp-up of the forcing function, which (by design) accelerates the plasma above and below $y=-L/4$ toward the x point. Since these converging flows also convect the magnetic field, they produce a buildup in both the plasma and magnetic pressures near the reconnection layer. This buildup is reflected by the early growth in the plasma density n and upstream reconnecting field B_d seen in Fig. 4(a). The growing pressure gradient and magnetic tension forces in the upstream region eventually become sufficient to compete with the applied force, at which point V_{in} starts to decrease and a new equilibrium state is approached. The decline in V_{in}, however, as well as the rapid growth of B_d, is halted at about $t\sim 28$ by the onset of rapid reconnection in the simulation. At this point the magnetic separatrix and out-of-plane current profile have fully opened up into the X-like geometry seen in Fig. 1, and the reconnection rate [dotted line in Fig. 4(b)], when properly adjusted for the magnitudes of B_d and n (discussed in detail below), has become comparable to the rates previously observed in large Δ', spontaneously reconnecting systems. The outflow and inflow velocities V_{out} and V_{in} shown in Fig. 4(a) are also consistent with these studies. The former has become comparable to the normalized upstream Alfvén speed $V_{Ad}=B_d/\sqrt{n}$, while the latter is consistent with the continuity relation

$$V_{\text{in}} \sim \frac{d}{D} V_{\text{out}} \sim \frac{1}{10} V_{Ad}, \tag{15}$$

where $d/D\sim 1/10$ is a rough estimate of the dissipation layer aspect ratio. The gradual rise of $10V_{\text{in}}$ relative to V_{out} seen in the figure following reconnection onset reflects a modest increase in the value of d/D beyond $1/10$, due to a moderate, gradual decrease in D.

As noted in the Introduction, the reconnection rate in the Alfvénic case is expected to scale as $E_r \propto B_d^2$. This result can be obtained by combining the estimate for V_{in} based on the $E\times B$ drift speed into the dissipation region, $V_{\text{in}}\sim cE_z/B_d$, with Eq. (15) and the estimate $V_{\text{out}}\sim V_{Ad}$

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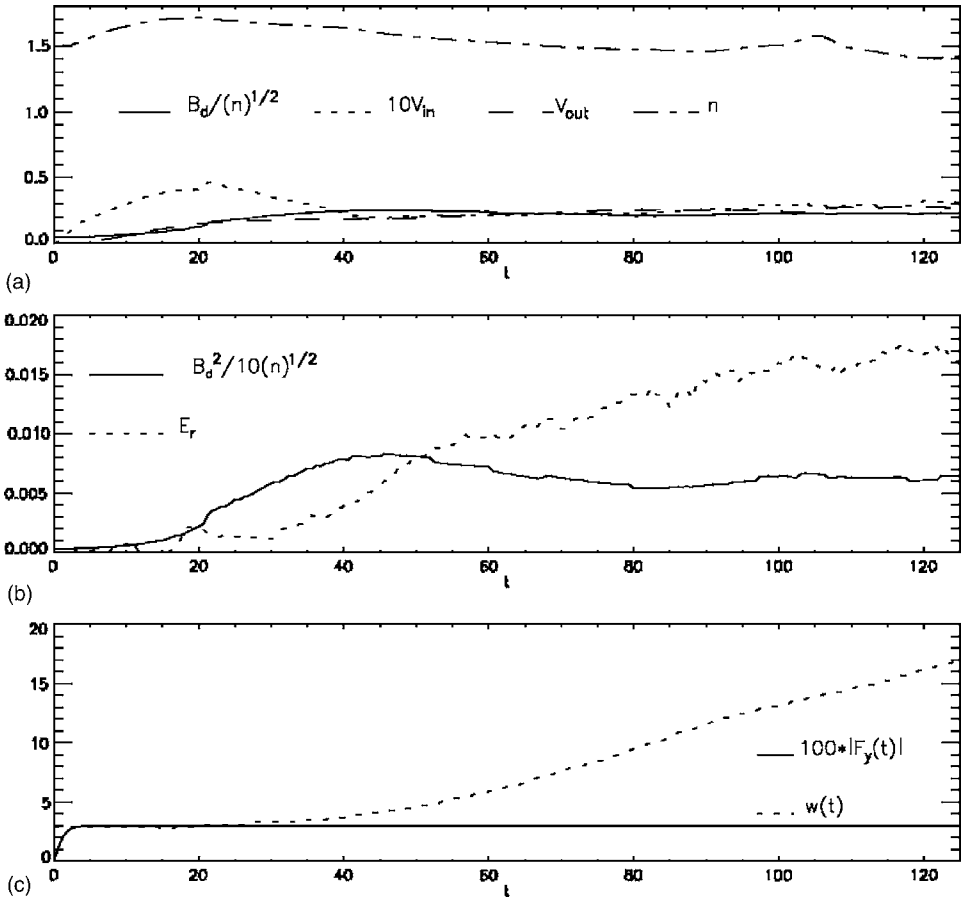


FIG. 5. Plateaued forcing with $F_\infty = 0.03$.

$$cE_z \sim V_{\text{in}} B_d \sim \frac{d}{D} V_{Ad} B_d = \frac{d}{D} \frac{B_d^2}{\sqrt{4\pi m_i n}}. \tag{16}$$

In normalized units this yields $E_z = (d/D) B_d^2/\sqrt{n} \sim (1/10) B_d^2/\sqrt{n}$. It is seen in Fig. 3(b) that this estimate is roughly consistent with the simulations. The gradual increase of E_z beyond $(1/10) B_d^2/\sqrt{n}$ is due to the increase of d/D beyond the value 1/10 noted earlier.

It was argued in Shay *et al.* (see Ref. 28), that the width of the current sheet had to fall to about 0.5 ion skin depth or less before the onset of fast (Alfvénic) reconnection. The sheet width in our simulations is initially on the order of the system size and, due to the forcing, falls to much smaller values. In the simulations without a guide field we observe fast reconnection to onset when the sheet width is reduced to about 0.5 or less (in physical units, given the density enhancement, this is typically about 0.7–0.8 ion skin depth). This result was also obtained in a simulation with a smaller electron mass ($m_e/m_i = 1/100$, $n_x \times n_y = 1024 \times 2048$), suggesting that the ion skin depth (rather than the electron skin depth) is indeed the most important factor. This threshold condition presumably arises from the importance of non-MHD effects (e.g., the Hall term in Ohm's law) in the fast reconnection process.

The fluctuations in V_{in} and B_d at late times ($t > 70$) are due to magnetosonic waves propagating in from adjacent x lines. These waves are generated when the adjoining outflows from the periodic array of x lines along $y = -L/4$ collide with each other.

Data from the next two simulations are shown in Figs. 4 and 5. These simulations each feature lower levels of forcing than the first simulation but the data are qualitatively similar. A comparison of these simulations shows that the final saturated island widths are nearly proportional to the final level of forcing, F_∞. The final values of F_∞ for these first three simulations are 0.11, 0.05, and 0.03, while the corresponding final saturated island widths are approximately 54, 26, and 17, respectively.

B. Weak forcing

The asymptotic level of forcing in the simulation shown in Fig. 6 is 1/5 that of the weakest forcing described above. At this level of forcing the data differ qualitatively from the first three simulations. First, observe that the duration of this run is much longer than those of the more strongly forced cases. Initially V_{in} increases as before, but in contrast to the previous runs, V_{in} then decreases all the way to zero. The variables V_{out} and E_r also decrease slowly back down to zero during the period $t > 200$. Additional simulations (not shown) indicate that the transition to Alfvénic reconnection occurs at a forcing level between $F_\infty = 0.005$ and $F_\infty = 0.0075$ for the parameters of this system. Below this level, the system is apparently able to reach a new quasiequilibrium state, halting the narrowing of the current sheet before the onset of non-MHD effects in the dissipation region.

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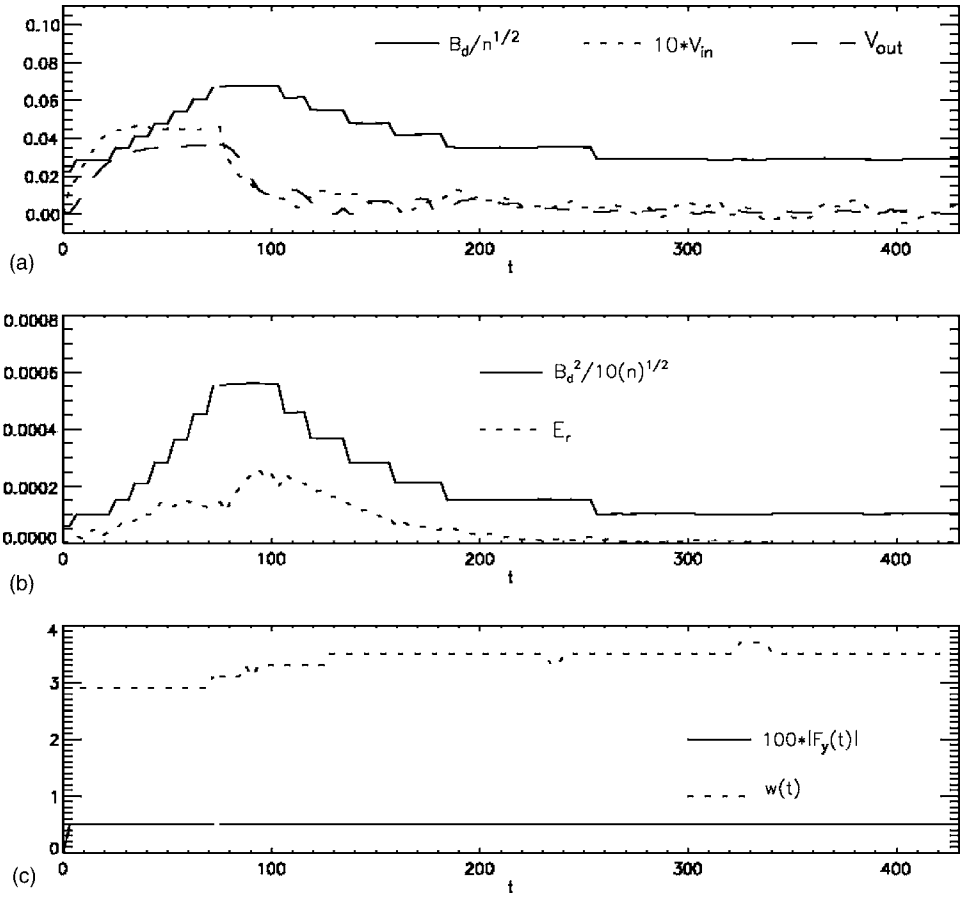


FIG. 6. Plateaued forcing with $F_\infty = 0.005$.

C. Linearly ramped forcing

In this simulation, the strength of the forcing is ramped linearly in time $[\Theta(t)=t/\tau_f]$. Time series of these data are shown in Fig. 7. The initial rate of ramping ($\tau_f=10$) is the same as in the earlier runs but the forcing never levels off. As before, the inflow increases initially due to ramp-up of the forcing, then decreases in response to the buildup of back pressure, and then increases again due to onset of fast reconnection. By the final time shown in the figure, the forcing function has reached a very high level, and has produced extremely steep gradients in the forcing region that are numerically challenging to resolve. As a result, this run was stopped at a somewhat earlier stage than the more weakly forced cases. It is included here because the extreme values of the upstream field produced in this simulation provide a useful test of the scaling laws discussed later.

D. Strong guide field

This simulation differs from the first (Fig. 4) only in that it includes a guide magnetic field, $B_{z0}=5.0B_{x0}$. Data are shown in Fig. 8. In this case B_d is measured at a distance $\Delta y=0.4$ upstream of the x point rather than $\Delta y=0.5$ as in the other simulations. This location was determined empirically by selecting a location just upstream of the point where the ion and electron inflow velocities separate. Although the initial hump in V_{in} is less pronounced, the behavior of B_d is similar to the $B_{z0}=0$ case. The peak reconnection rate is reduced by about a factor of 2 by the presence of $B_{z0}\neq 0$, and consistent with this, the current sheet opens to an angle of about half that seen in Fig. 1. This slowdown is consistent with behavior noted in spontaneously reconnecting systems (see, for example, Refs. 34 and 35).

E. Varying width of the forced region

To explore the dependence of the reconnection process on the width w_x of the forcing function along x, we present data in which w_x has been decreased (Fig. 9) and increased (Fig. 10) by a factor of $\sqrt{2}$. The other parameters, for example the forcing level $F_\infty=0.05$, are the same as those in the second simulation (Fig. 4). We therefore have three simulations with forcing widths of approximately 7, 10, and 14 d_i in Figs. 4, 9, and 10, respectively. Comparing these three simulations in order of increasing width, we find that the magnetic island width w levels off at 22, 26, and $30d_i$. The upstream magnetic field in each of these cases levels off at approximately 0.4, 0.5, and 0.6 B_{x0}, respectively.

In Fig. 11(a), we plot time series data of the reconnection rate E_r for the three simulations with forcing functions of varying width. We see that the reconnection rate at late times varies directly with the width of the forcing function w_x, i.e., wider forcing functions ultimately produce higher rates of reconnection. This increase in the raw values E_r, however, is more than accounted for by the increase in B_d with w_x shown in Fig. 11(b). Adjusting E_r for the variation of B_d as in Fig. 11(c), one sees that the wider forcing widths in fact have somewhat *lower* adjusted rates of reconnection. This is per-

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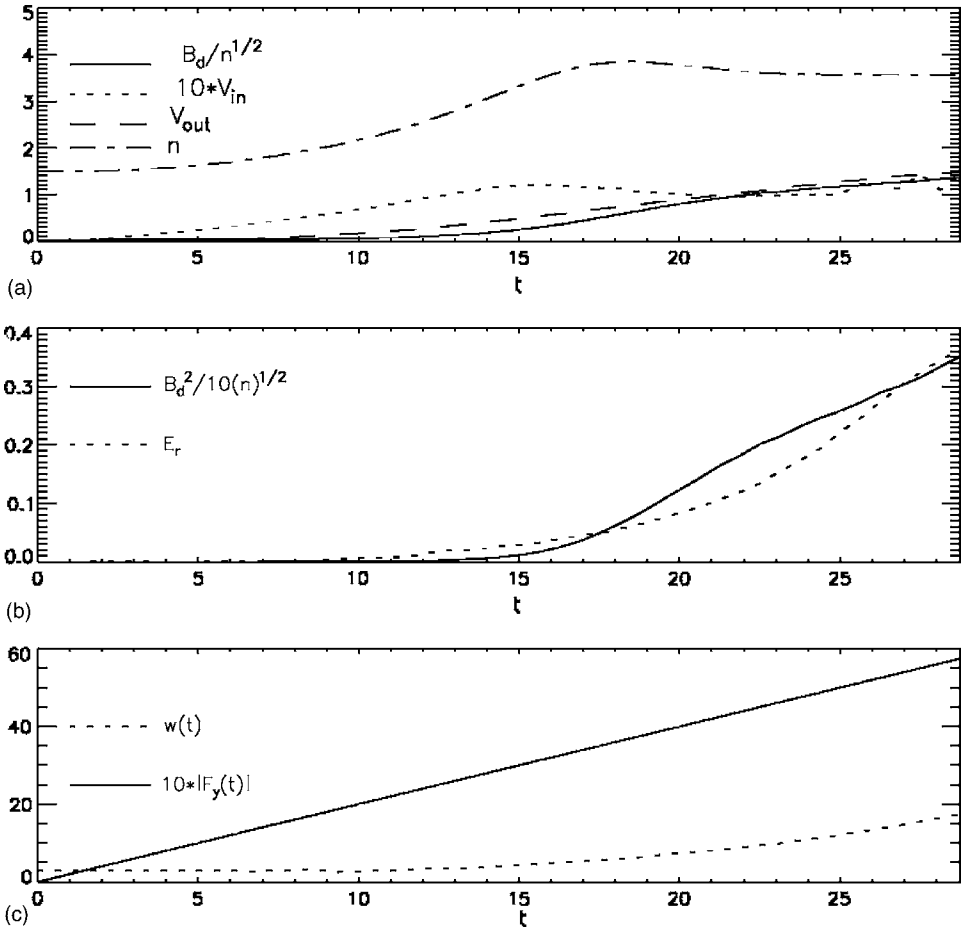


FIG. 7. Time series data from continuously ramped simulation.

haps due to a tendency of wider forcing profiles to "pinch" the reconnection layer. Further study is necessary to fully explore the dependence of the reconnection rate on the forcing profile.

As noted earlier, the forcing function is antisymmetric in the vertical (y) direction about the reconnection layer and therefore vanishes at $y=-L/4$. In all the simulations presented here, it then increases/decreases monotonically to the constant values $\pm F_\infty$ over a distance of about $2w_y \sim 5d_i$ away from the layer in the upstream regions. As w_y is increased substantially beyond this value for a fixed level of forcing, thereby moving the forced zone away from the dissipation layer, the compression of the flux surfaces near the dissipation layer due to the forcing is spread out over a larger region. This weakens the impact of the forcing on the reconnection zone and therefore necessitates a further strengthening of the forcing level in order to achieve a narrow current sheet. As mentioned in the case of the continuously ramped simulation, however, very large forcing levels tend to generate steep gradients within the forced zone (that is, far upstream from the reconnecting region) due to the evacuation of plasma and magnetic flux from the forced regions, making the simulations progressively more challenging. We have therefore not presented a scaling study with respect to the parameter w_y in this article.

F. Lower β simulation

In this simulation, shown in Fig. 12, the total temperature has been lowered from 1.0 to 0.5 (corresponding to an upstream $\beta=1$). All other parameters are the same as in the simulation shown in Fig. 3. One sees that the peak reconnection rate is lower by just over a factor of 2 (compare with Fig. 3), a finding that is also consistent with previous studies of spontaneously reconnecting systems (see, e.g., Ref. 35). The total island growth is also much less than that observed in Fig. 3. However, B_d still levels off at approximately the same level as that observed in the higher β case.

IV. DISCUSSION

The validity of the simple scaling laws $V_{\text{out}} \sim B_d/\sqrt{4\pi m_i n}$ and $E_r \propto B_d^2/\sqrt{n}$ [Eq. (16)] in our simulations can be better illustrated by pooling together the results of the various simulations discussed thus far. In Figs. 13 and 14, data from all of the simulations at fixed $T=1$, $w_x=10$, $B_{z0}=0$, as well as a fourth simulation (not previously discussed) with $F_\infty=0.14$, are plotted on the same axes with time as a parameter. Each data point represents a measurement taken from the simulations at a particular instant of time. The time separation between measurements, which start at the time of the first local minimum of V_{in} and end at the time that magnetosonic waves from adjacent x lines begin to strongly af-

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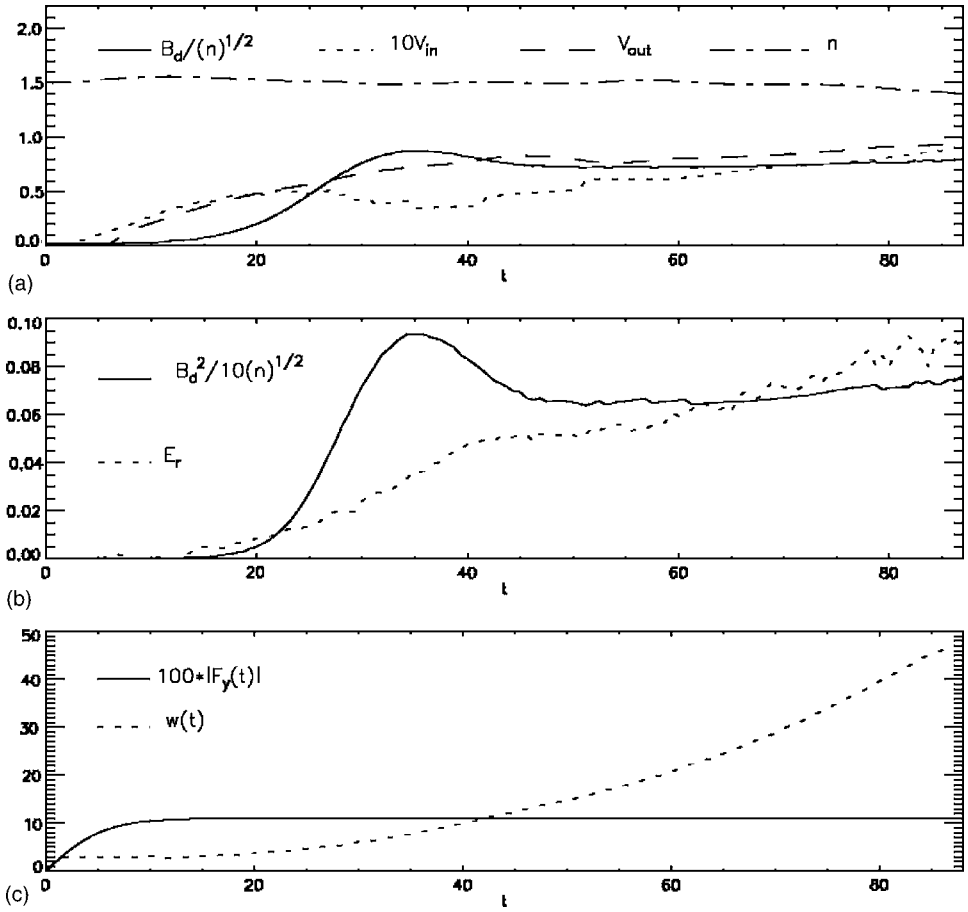


FIG. 8. Time series data from simulation with a guide field of $B_{z0}=5.0$.

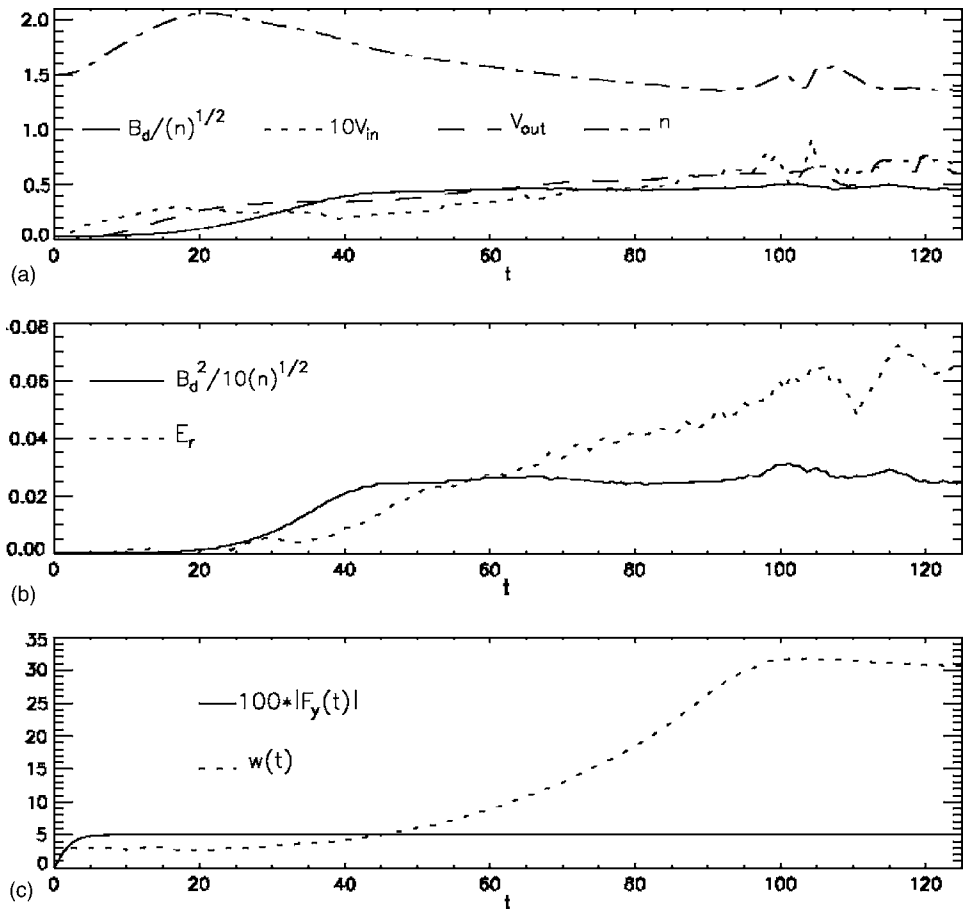


FIG. 9. Time series data from simulation with wider forcing function, $w_x=10\sqrt{2}d_i$, $F_\infty=0.055$.

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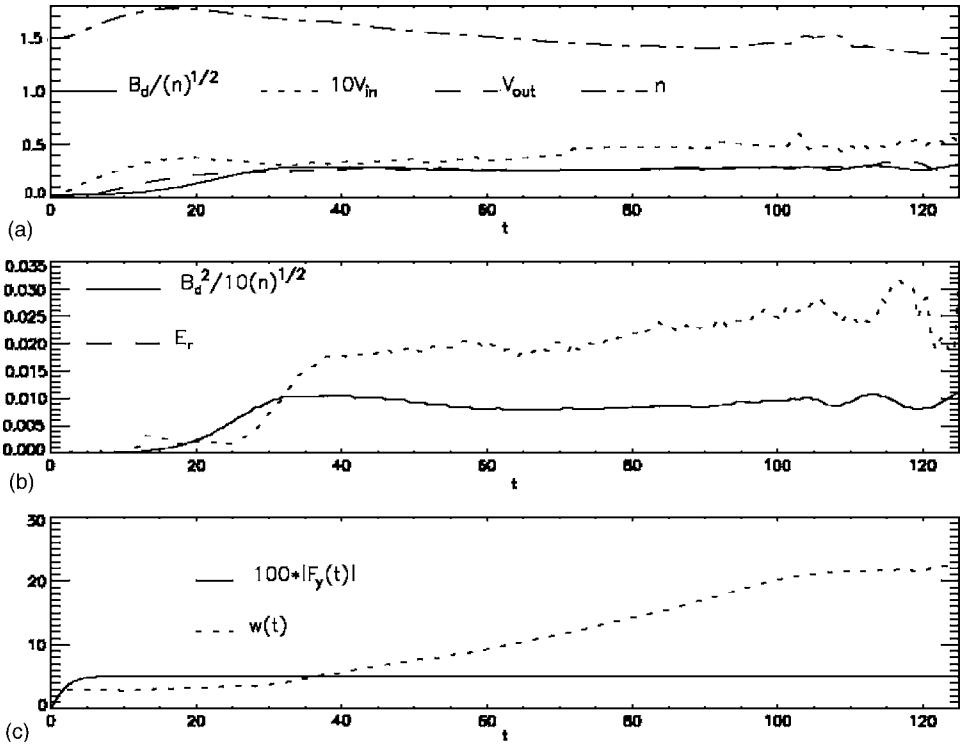


FIG. 10. Time series data from simulation with narrower forcing function, $w_x=(10/\sqrt{2})d_i$, $F_\infty=0.055$.

fect the dissipation region, is the same for each simulation. To further probe the scalings for larger values of B_d, data from the simulation with linearly ramped forcing (seen as the cluster of points in the upper right-hand corner of each plot) are also included. We also account in the plots for variations in the density in the dissipation region, which tend to increase with the level of forcing and have a notable impact on the scaling results. The dotted line of slope unity in Fig. 13, plotted for reference, represents an outflow velocity equal to the upstream Alfvén speed based on B_d. Likewise, the dotted line in Fig. 14 has a slope of $1/\sqrt{10}$ and represents the scaling we would expect from Eq. (16), assuming the aspect ratio of the dissipation region in the simulations remains roughly fixed at $d/D=1/10$. These figures suggest that in the system explored here the simple scaling laws for V_{out} and E_r

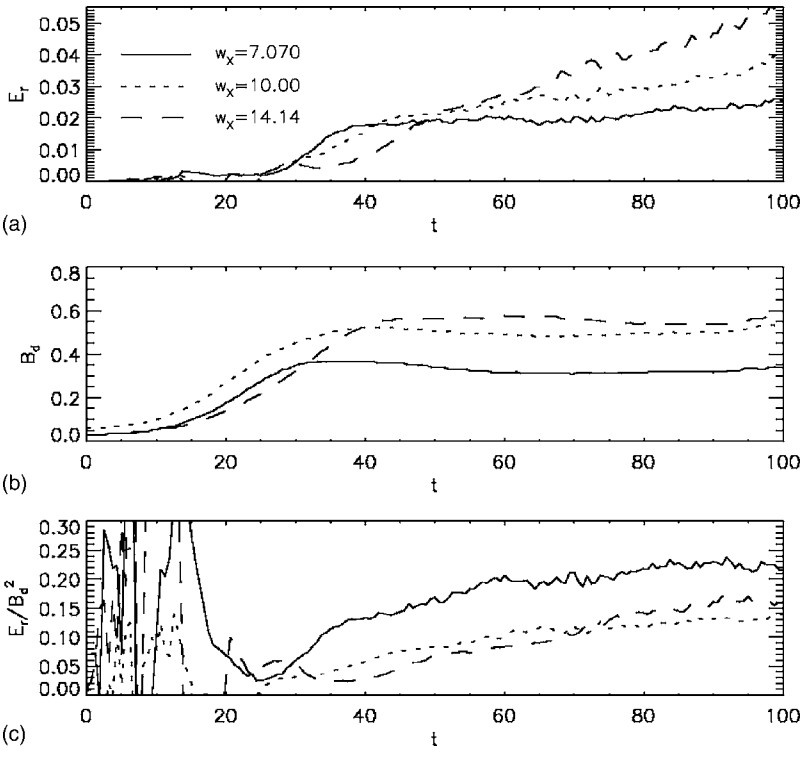


FIG. 11. Scaling of E_r with w_x.

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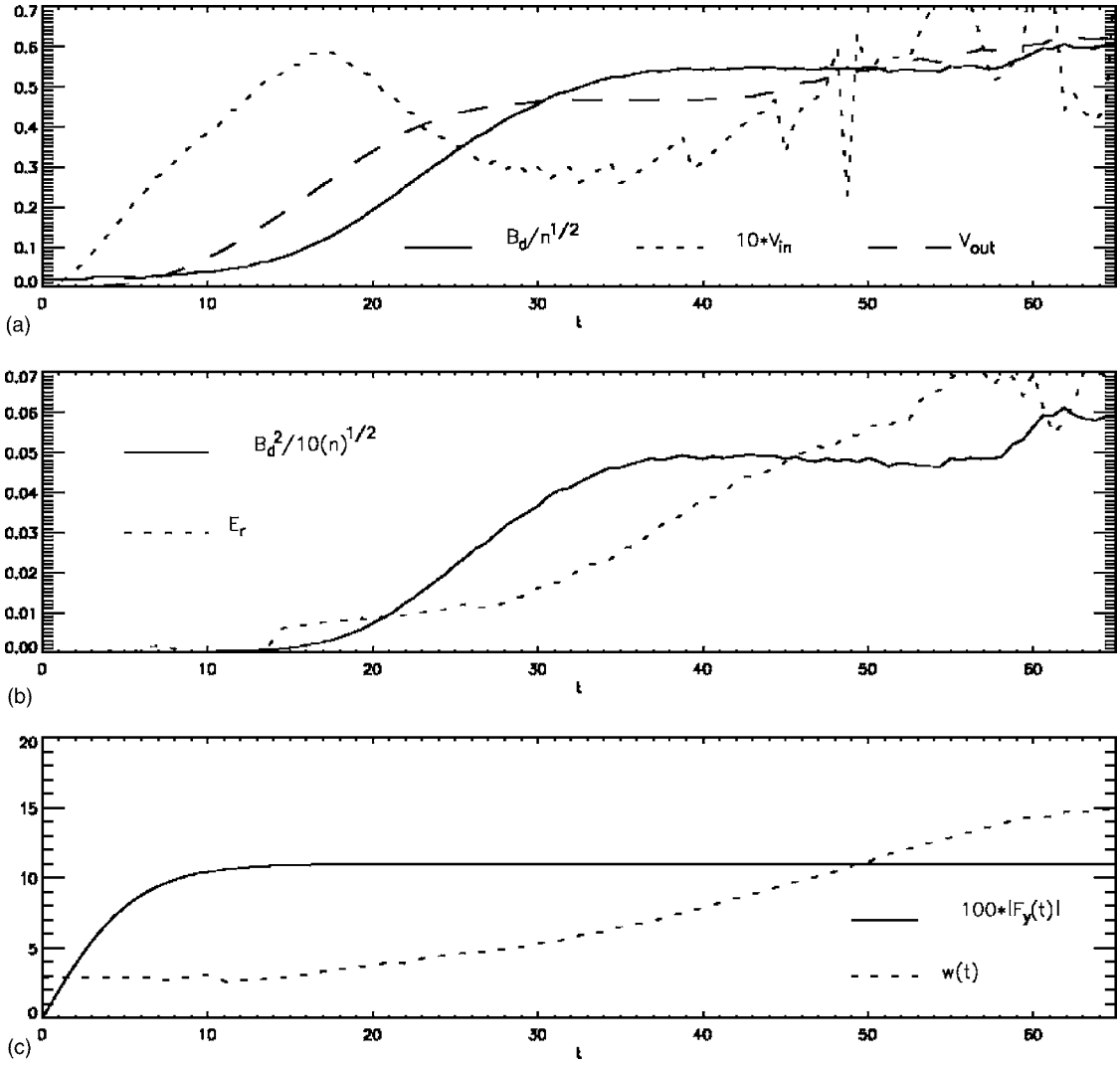


FIG. 12. Time series data from simulation with lower β.

based on the upstream field are indeed reasonably well satisfied when the forcing is made sufficiently strong.

In an effort to more closely compare our simulation results to those of spontaneously reconnecting systems with large Δ', in the figures we also plot data from a periodic nonforced simulation of the same box size, grid size, upstream plasma β, and electron mass. Similar to the configuration of the GEM challenge system, the initial equilibrium in this simulation has a narrow current sheet of width $\sim d_i$ and a constant magnetic field outside the sheets. It is seen that the relative outflow velocities and reconnection rates in our forced system are $\sim$50% lower than in the spontaneously reconnecting, large-Δ' case. A similar effect was noted in the recent forced reconnection study of Birn *et al.* (see Ref. 36).

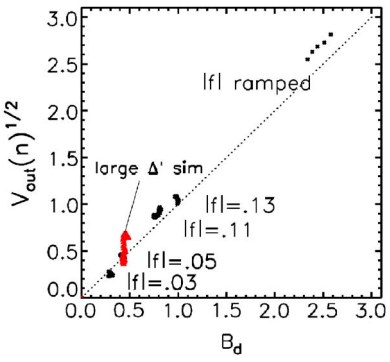


FIG. 13. (Color online). Scaling of V_{out} with B_d.

V. CONCLUSIONS

We have examined the scaling of forced magnetic reconnection in a two-dimensional periodic system using two-fluid simulations with finite electron inertia. The forcing in the simulations was driven by a spatially localized forcing function inside the computational domain. We investigated the dependence of the reconnection process on the forcing level, the width of the forced region, the plasma-β, and the guide magnetic field strength.

As the forcing is turned on in the simulations, plasma and magnetic flux are driven toward the x point, causing a rise in the inflow velocity V_{in}, the reconnecting magnetic field B_d, and plasma pressure in the vicinity of the dissipation region, as well as a narrowing of the current sheet. This plasma inflow is eventually checked by the growth of the

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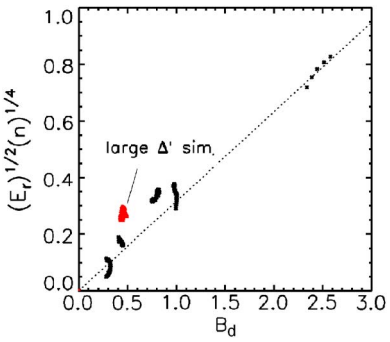


FIG. 14. (Color online). Scaling of $\sqrt{E_r}$ with B_d (values of $|f|$ are the same as in Fig. 12).

total pressure and magnetic tension forces near the layer, which act to restore the plasma to equilibrium in the presence of the applied force. If the strength of the forcing is sufficient, however, the onset of rapid reconnection in the simulation causes the inflow V_{in} to start increasing again in time. The upstream field B_d remains fairly constant during the reconnection phase, indicating that the input of magnetic flux into the dissipation region is approximately in balance with the evacuation rate due to reconnection.

Based on the simulations included here and consistent with past work (see, for example, Ref. 28), it appears to be a necessary condition for the onset of fast magnetic reconnection that the current sheet be narrowed to a width of somewhat less than 1 ion skin depth (approximately $0.7-0.8d_i$). If the magnitude of the applied force is insufficient to overcome back-pressure in the dissipation region and thereby sufficiently reduce the width of the current sheet, Alfvénic reconnection is not observed to occur.

By varying the strength of the forcing, we found that the behavior of the reconnection rate and plasma flows during the fast reconnection phase of the simulations are in rough agreement with the scalings obtained in large-Δ' spontaneously reconnecting systems.[37] For example, the outflow velocity from the dissipation region is comparable to the Alfvén speed based on B_d while the reconnection rate scales like B_d^2. Also consistent with the behavior of spontaneously reconnecting systems in the parameter regime considered here, the lowering of the plasma β or the addition of a strong guide field ($5.0B_{x0}$) modestly reduces the peak magnetic reconnection rate and the opening angle of the current sheet.

ACKNOWLEDGMENT

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